

Thermodynamics, Fluids and Heat Transfer Report

Joseph Bradshaw ME2TF1

Part 1: Refrigeration Lab

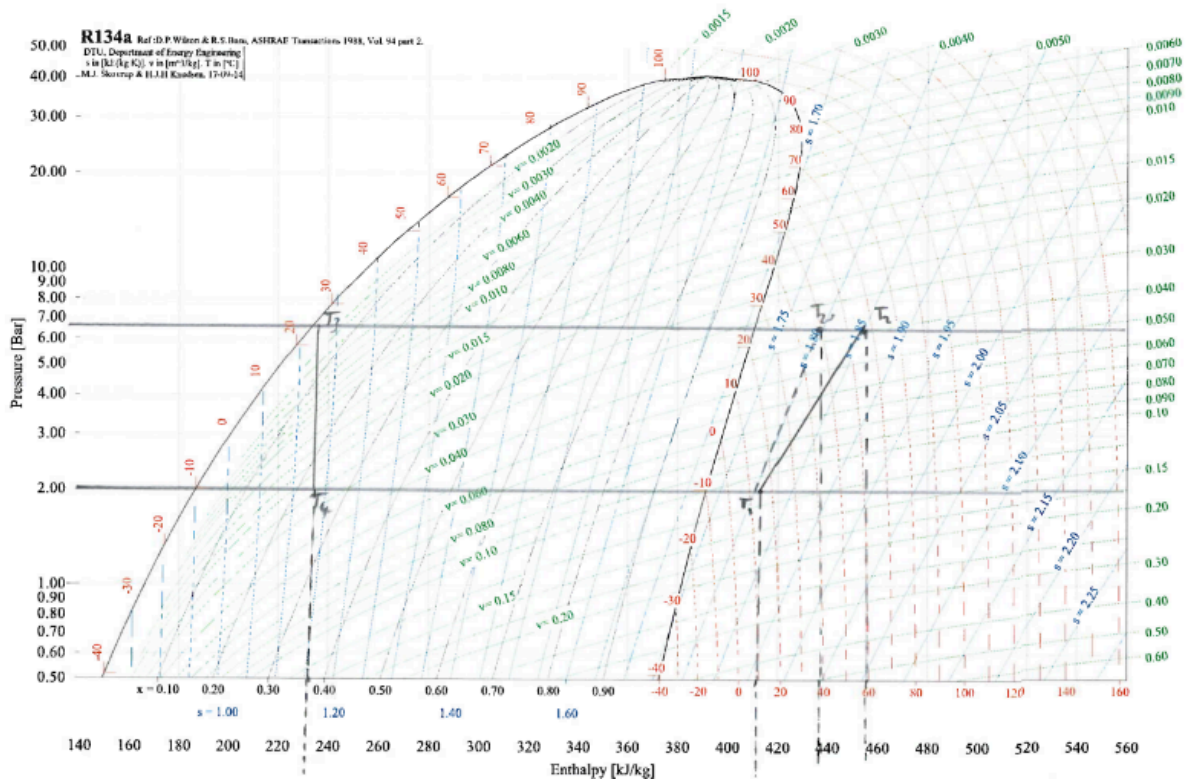
1.1 Objectives

Investigate the performance of a vapour compression refrigerator with varying condenser pressures and temperatures.

1.2 Results Analysis

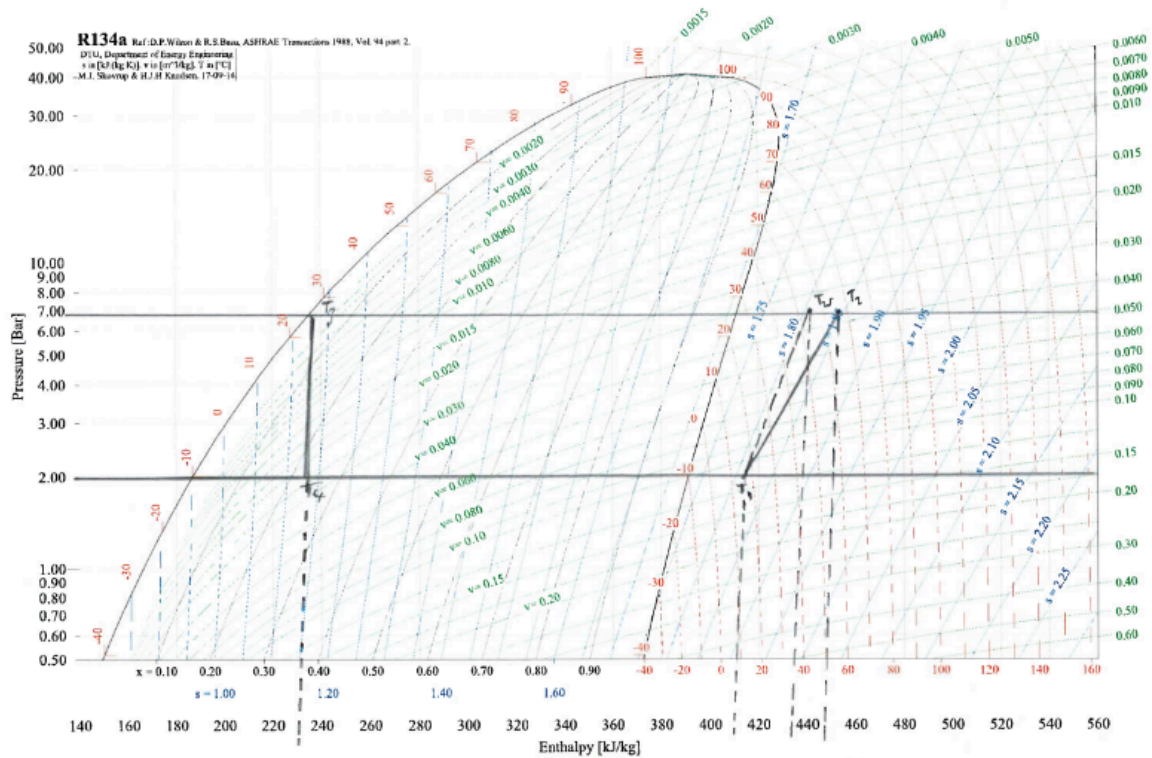
Test 1

①



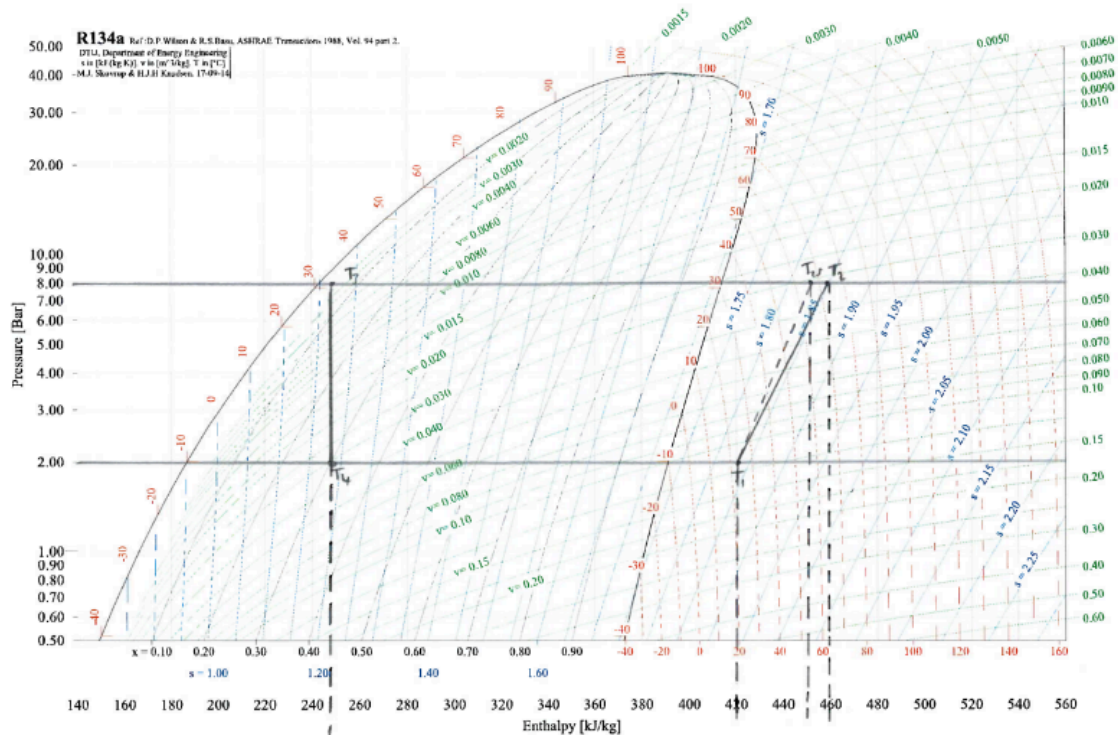
Test 2

③

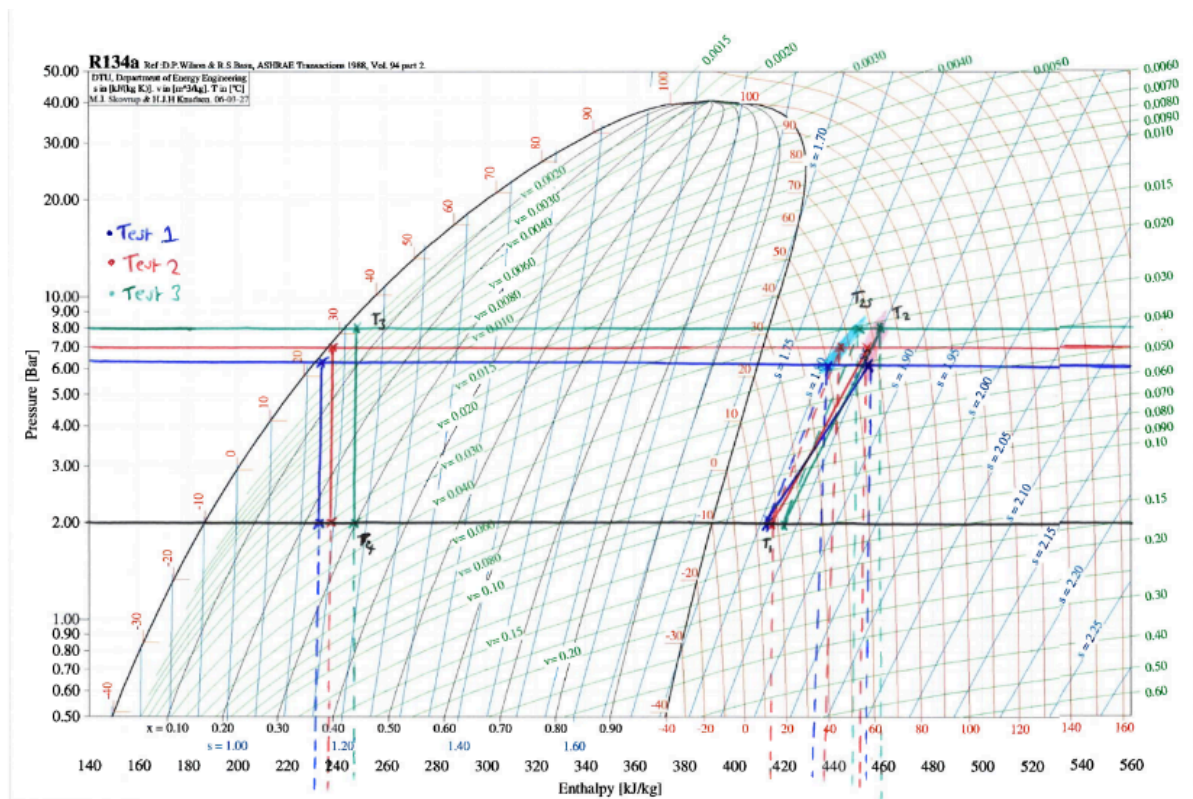


Test 3

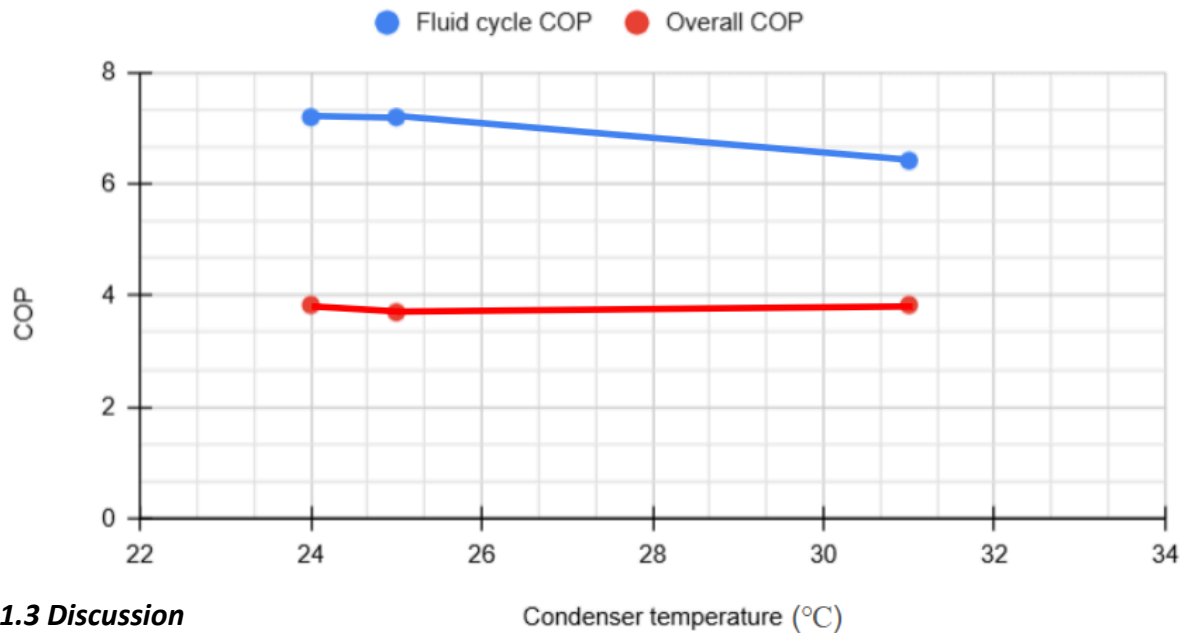
③



Test 1/2/3 combined



Fluid cycle COP and Overall COP



1.3 Discussion

Throughout, condenser pressure increases. The compressor is working across a different pressure ratio and operating point- the P-h graphs above show the actual compression line getting closer to the ideal isentropic line, meaning less irreversibility/kg, increasing compressor isentropic efficiency. The compressor is working closer to ideal isentropic compression in Tests 2/3 than in Test 1, potentially caused by changes in condenser pressure/temperature.

My calculated COP values are much lower than Carnot, around 3.8 compared to 7.5. This is because Carnot assumes perfect conditions that don't exist in this experiment [1]- in reality the experiment has non-isentropic compression, with pressure losses throughout the evaporator/condenser and valves.

Overall COP doesn't fall even though T_c rises- this is because even though ideal COP drops as T_c increases, the measured COP stays around 3.7/3.8. This is caused by other factors changing e.g. mass flow rate and enthalpy [1], meaning the COP can look flat due to multiple effects competing. From highest to lowest, the COP values should be as follows:

$$\text{COP}_{\text{Carnot}} \geq \text{COP}_{\text{isentropic cycle}} > \text{COP}_{\text{shaft/fluid}} > \text{COP}_{\text{overall}}$$

My results follow this order, meaning each source of irreversibility is being captured successfully by the experiment and calculations.

1.4 Conclusion

Increasing condenser pressure and saturation temperature caused a higher compressor discharge temperature and pressure ratio, increasing the specific compression work required. This was visible on the P–h diagrams, where the enthalpy rise during compression increased as condenser pressure increased. The compressor isentropic efficiency improved from approximately 56% to 75%, indicating that the compressor operated closer to its optimal efficiency at higher condenser pressures. Despite this improvement, the overall system COP remained relatively constant at ≈ 3.7 – 3.8 . This showed that increases in compressor efficiency were balanced by increased compression work and significant motor/electrical losses [2], limiting improvements in overall system performance.

The experimental COP was significantly lower than the ideal Carnot COP values, as expected. The decrease from Carnot COP to isentropic cycle COP, shaft-based COP, and overall COP confirmed these losses were cumulative and proved the validity of the results. Overall, the experiment showed that increasing condenser pressure and temperature affects the thermodynamic efficiency potential of a vapour compression refrigerator, even when compressor efficiency improves. The results highlight the importance of maintaining low condenser temperatures in practical refrigeration systems to maximise COP and overall performance.

1.5 References

- [1]- ([Thermodynamics: An Engineering Approach](#)), 9th ed., McGraw-Hill Education.
Yunus A. Çengel & Michael A. Boles, (2020).
- [2]- ([Principles of Refrigeration](#)), 5th ed., Prentice Hall.
Roy J. Dossat, Thomas J. Horan (2002).

Part 2: Pump systems Lab

2.1 Objectives

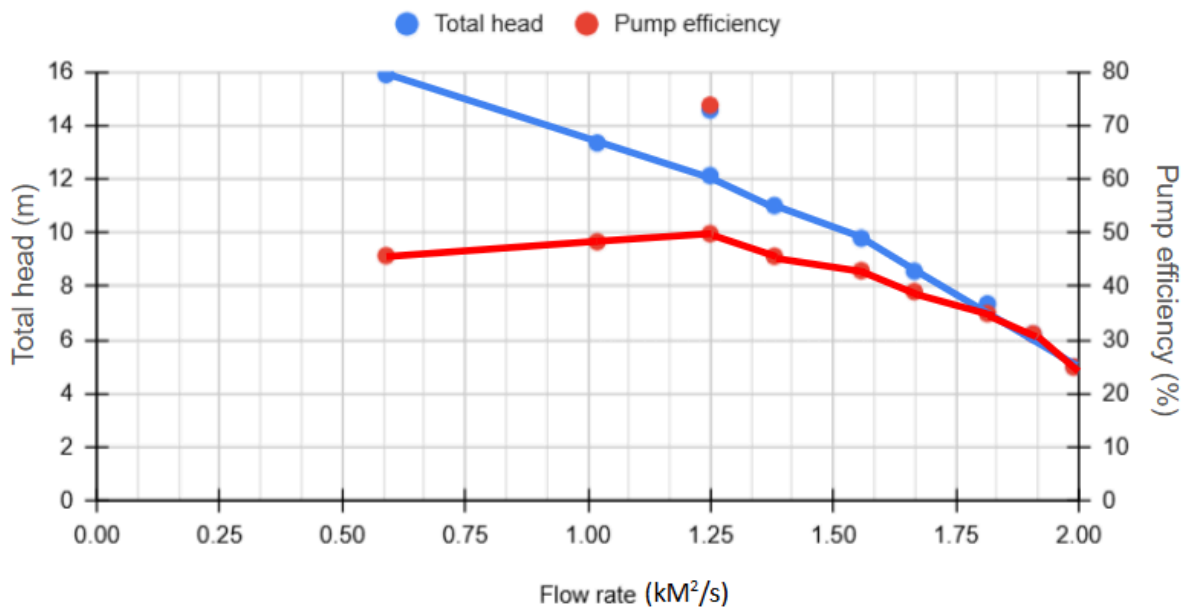
-Obtain a set of head-flow curves for a centrifugal pump operating at a constant speed, and to relate those parameters to pump power and efficiency

-Demonstrate the operating characteristics of two pumps running in a series pumping installation, the operating characteristics of two pumps running in a parallel pumping installation and the effect of cavitation on pump performance

2.2 Results Analysis

Constant speed operation (3000 rev/min)

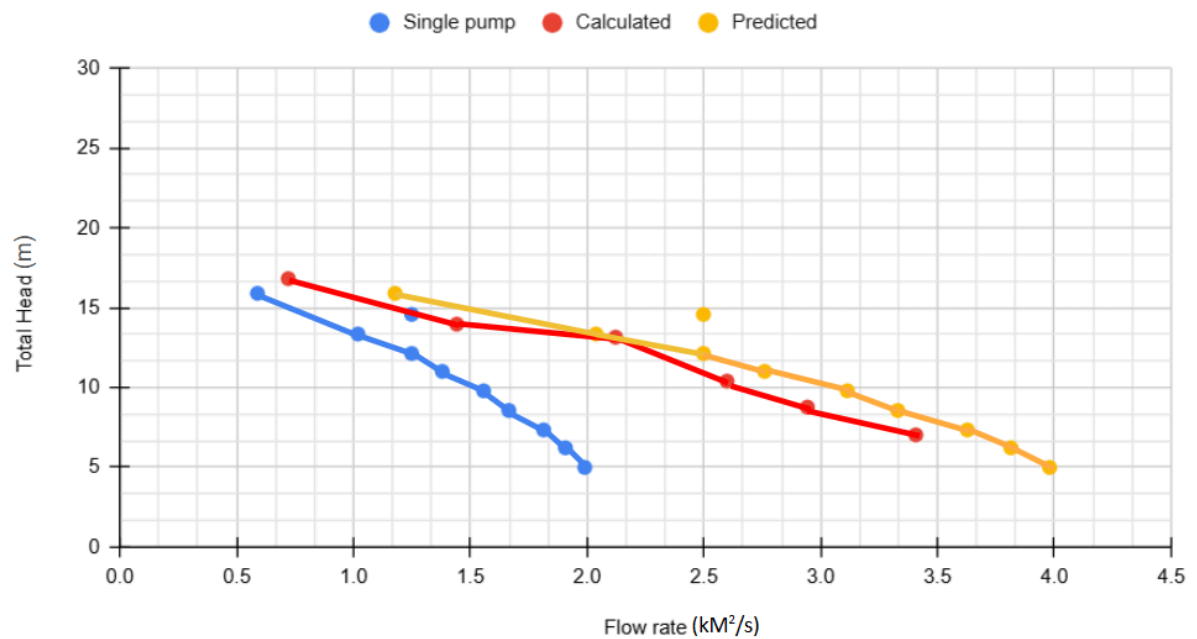
Total head & Pump efficiency against Flow rate



Flow rate (km ² /s)	Total head (m)	Pump efficiency (%)
1.99	4.99	24.9
1.907	6.22	31.1
1.814	7.34	34.9
1.665	8.56	39.0
1.557	9.79	42.9
1.380	11.01	45.6
1.249	12.13	49.8
1.019	13.35	48.3
1.249	14.58	73.8
0.589	15.90	45.7

Parallel pump operation

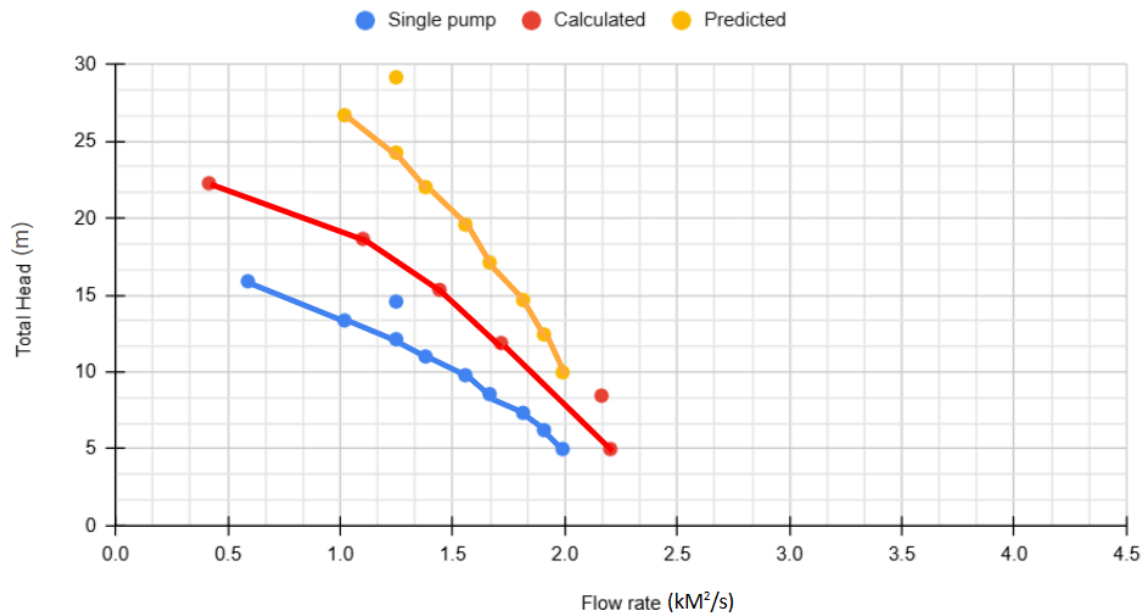
Parallel pump operation



Flow rate	Total head (calculated)	Flow rate	Flow rate (prediction)	Constant pressure head
3.407	7.03	1.99	3.98	4.99
2.943	8.77	1.907	3.814	6.22
2.599	10.40	1.814	3.628	7.34
2.122	13.15	1.665	3.33	8.56
1.442	13.97	1.557	3.114	9.79
0.721	16.82	1.38	2.76	11.01
		1.249	2.498	12.13
		1.019	2.038	13.35
		1.249	2.498	14.58
		0.589	1.178	15.9

Series pump operation

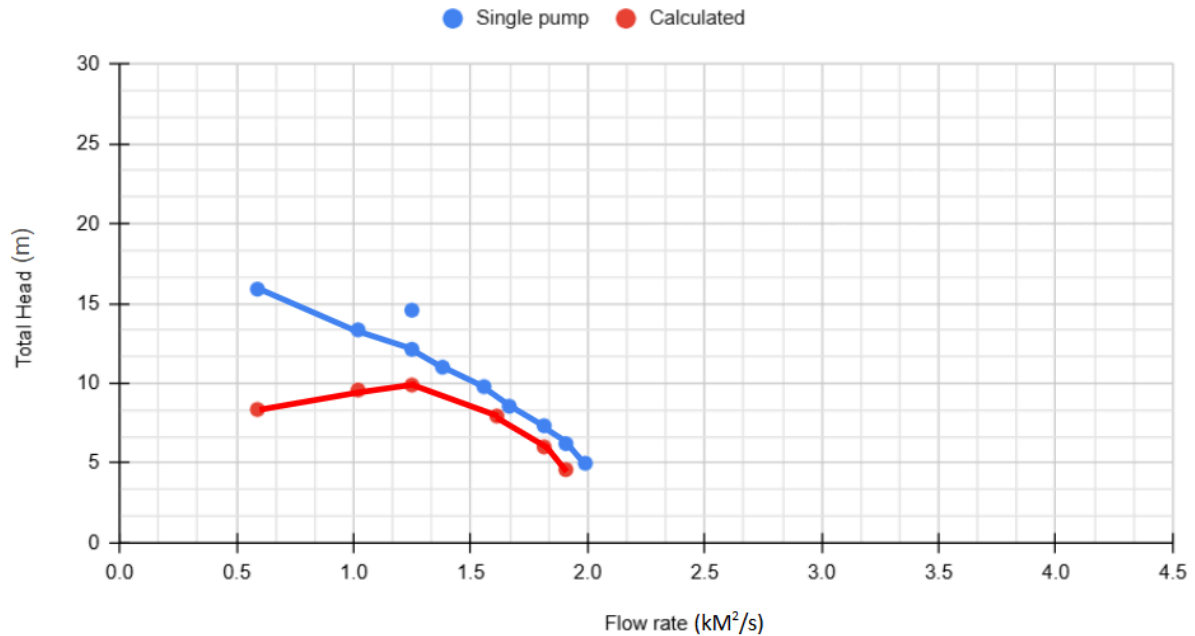
Series pump operation



Flow rate	Total head (calculated)	Flow rate	Total Head (prediction)	Constant pressure head
2.202	4.99	1.99	9.98	4.99
2.163	8.46	1.907	12.44	6.22
1.716	11.88	1.814	14.68	7.34
1.442	15.34	1.665	17.12	8.56
1.101	18.65	1.557	19.58	9.79
0.416	22.27	1.38	22.02	11.01
		1.249	24.26	12.13
		1.019	26.7	13.35
		1.249	29.16	14.58
		0.589	31.8	15.9

Suction performance test

Suction Performance Test & Cavitation Demonstration



Flow rate	Total head (calculated)	Flow rate	Single pump head
1.907	4.59	1.99	4.99
1.814	6.01	1.907	6.22
1.612	7.95	1.814	7.34
1.249	9.89	1.665	8.56
1.019	9.58	1.557	9.79
0.589	8.36	1.38	11.01
		1.249	12.13
		1.019	13.35
		1.249	14.58
		0.589	15.9

In the constant speed graph, total head decreases as flow rate increases. As flow increases, the internal hydraulic losses rise with velocity- more energy is then dissipated rather than converted into pressure head [1]. Pump efficiency peaks at around 50%- at low flow, internal recirculation and leakage become more dominant, and at high flow, friction and

shock/incidence losses increase. Both of these increase losses and reduce efficiency.

For two identical pumps operating in parallel, the total head at any point is equal to a single pump, while the total flow rate is the sum of the individual pump flow rates [3]. Therefore, the predicted parallel pump curve was calculated by doubling the flow rate at each head value from the single-pump curve. The resulting head–flow curve retains the same shape as the single-pump but is shifted to higher flow rates, showing the parallel pump arrangement increases capacity more than it increases head [2].

For two identical pumps in series, the flow rate is the same while the total head is the sum of the individual head [3]. Therefore, the predicted series curve is obtained by doubling the head at each flow rate from the single-pump curve. Similar to the parallel pump setup, the actual results are less than predicted due to system losses [2], except this time series setups increase achievable head substantially.

The bubble observations show that worse cavitation causes reduced performance- when local pressure falls below vapour pressure, vapour bubbles form at the impeller inlet [1]. These bubbles occupy volume and disrupt the liquid flow, reducing the energy transfer to the fluid. This results in lower head and therefore higher pump efficiency.

2.4 Conclusion

The head–flow curve showed the expected decrease in total head with increasing flow rate, displaying the effect of frictional and hydraulic losses. The predicted curves assumed the pumps were identical with perfect speed control and no losses- this was not the case.

Two pumps in series produced a higher total head at a given flow rate, demonstrating that this setup is effective for applications requiring increased pressure rise. Parallel pumps increased the achievable flow rate at approximately the same head, showing that parallel operation is suitable for increasing system capacity [1]. The cavitation test demonstrated that reducing suction pressure leads to bubble formation and a significant reduction in pump head/efficiency, highlighting the importance of good suction conditions in centrifugal pump operation.

2.5 References

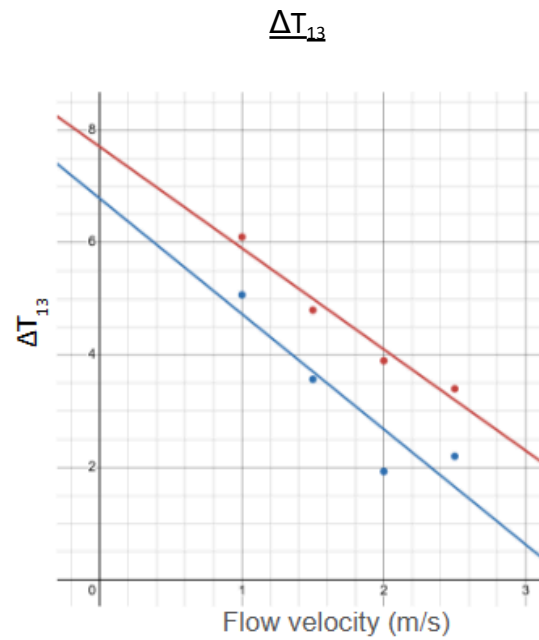
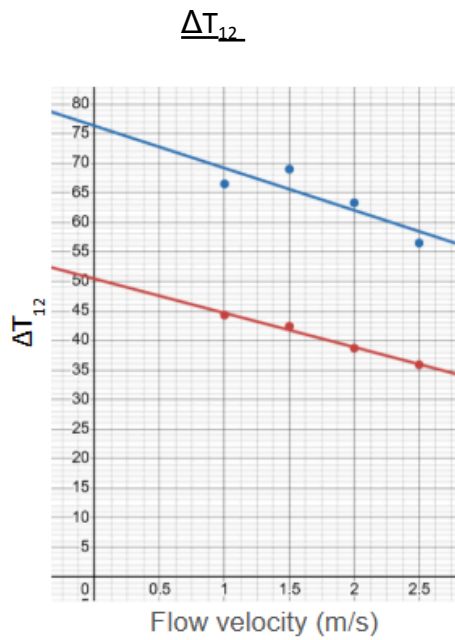
- [1]- ([Introduction to Fluid Mechanics](#)), 9th ed., Wiley.
Philip J. Pritchard & John W. Mitchell (2020)
- [2]- ([Pump handbook](#)), 4th ed., McGraw-Hill.
Igor J. Karassik, Joseph P. Messina, Paul Cooper & Charles C. Heald
- [3]- ([Pumps and Systems](#)), PDF 2024

Part 3: Heat transfer Lab (remove if ME2011)

3.1 Objectives

To understand the effect of flow velocity on the convection heat transfer in extended surfaces.

3.2 Results Analysis



	ΔT_{12}			ΔT_{13}	
Flow velocity	Pinned surface	Finned surface	Flow velocity	Pinned surface	Finned surface
1	44.3	66.5	1	6.1	5.07
1.5	42.4	69.0	1.5	4.8	3.57
2	38.7	63.3	2	3.9	1.93
2.5	35.9	56.5	2.5	3.4	2.20

3.3 Discussion

As velocity increases, ΔT_{12} decreases- increasing the velocity makes the thermal boundary layer thinner, increasing near wall mixing. Near wall mixing is how well the

fluid moves [2] and exchanges momentum/energy close to the heated surface. Increasing this increases the convection coefficient- with the heater power being held at 50W, the system needs less temperature driving force [2] to reject the same heat. This causes surface temperature to fall at the inlet, decreasing ΔT_{12} .

ΔT_{13} also decreases with velocity. As velocity increases, mass flow rate increases roughly in proportion to velocity- each kilogram of air that flows through the duct spends less time there and picks up less temperature rise, causing ΔT_{13} to fall.

In relation to the Nusselt number, this is a way of measuring how much better heat transfer is to pure conduction. [1] It's calculated through using the Reynolds number and Prandtl number of a fluid- the formula for the Reynolds number is: $Re = \frac{uL}{\nu}$. The Prandtl number is roughly constant throughout the experiment [1], meaning the convection is mainly affected by the Reynolds number. From the equation, this is directly dependent on velocity, meaning as velocity increases near wall mixing also increases.

3.4 Conclusion

Increasing flow velocity consistently reduced the surface/air temperature difference, demonstrating improved convective heat transfer due to increased Reynolds and Nusselt numbers and convection heat transfer coefficient [3]. Although the air temperature rise across the test section decreased with increasing velocity, this was caused by the higher mass flow rate rather than reduced heat transfer. When comparing surface geometries I found the pinned surface generally achieved better heat transfer performance than the finned surface, highlighting the importance of near-wall mixing and surface area. Overall, the results confirm that higher flow velocities significantly improve convection heat transfer from extended surfaces.

3.5 References

- [1]- ([Fundamentals of Mass and Heat Transfer](#)), 8th ed., Wiley.
Theodore L. Bergman & Adrienne S. Lavine (2017)
- [2]- ([Convective Heat and Mass Transfer](#)), 4th ed., McGraw-Hill.
William M. Kays, Michael E. Crawford & Bernhard Weigand (2005)
- [3]- ([Introduction to Heat Transfer](#)), 6th ed., Wiley.
Theodore L. Bergman, Adrienne S. Lavine, Frank P. Incropera & David P. Dewitt (2011)

Appendix A (Refrigeration Lab)

A.1 Table of Collected/Observed/Supplied results

In this section, present observed or supplied data. Present them clearly and professionally.

			Test number		
Description			1	2	3
<u>Temperatures</u>					
Compressor suction	T_1	$^{\circ}\text{C}$	14.7	18.7	22.3
Compressor delivery	T_2	$^{\circ}\text{C}$	66.1	67.0	72.5
Condenser exit	T_3	$^{\circ}\text{C}$	24.2	25.6	32.4
Evaporator inlet	T_4	$^{\circ}\text{C}$	-12.5	-11.6	-10.8
Cooling water inlet	T_5	$^{\circ}\text{C}$	20.0	20.0	20.4
Cooling water outlet	T_6	$^{\circ}\text{C}$	23.5	25.3	31.9
<u>Pressures</u>					
Evaporator (compressor suction)	P_e	kN/m^2	100	100	100
Condenser (compressor delivery)	P_c	kN/m^2	550	600	700
<u>Electrical</u>					
Evaporator load	Q_{ee}	W	672	673	682
Motor input	W_{me}	W	470	486	502
<u>Miscellaneous</u>					
Refrigerant Flow Rate	m_r	g/s	10	10	11
Cooling Water Flow Rate	m_w	g/s	50	30	10
Compressor speed	N_c	rev/min	847	846	844
Motor torque arm force	F_s	N	12	12	12
Atmospheric pressure	P_a	kN/m^2	Take as 100kN/m^2		

A.2 Sample Calculation

$h_1, h_2, h_3, h_4, h_{12}$ - off p-h graph
 Q_{ee} - from results
 $Q_{er} = m_r (h_1 - h_4) = 10 (410 - 230) = 1800$

$$\text{COP}_{\text{isentropic}} = \frac{Q_{\text{ev}}}{W_{\text{fi}}} = \frac{1800}{250} = 7.2$$

$$\text{COP}_{\text{carnot}, T_c} = \frac{T_e}{T_c - T_e} = \frac{263}{297 - 263} = 7.74$$

$$\text{COP}_{\text{carnot}, T_s} = \frac{T_e}{T_s - T_e} = \frac{263}{293 - 263} = 8.77$$

A.3 Table of Calculated Results

			Test number			
Description			1	2	3	4
<u>Enthalpy</u>						
Compressor suction	h_1	kJ/kg	410	410	420	
Compressor delivery (actual)	h_2	kJ/kg	455	445	460	
Compressor delivery (assuming isentropic compression)	h_{2s}	kJ/kg	435	435	450	
Condenser exit	h_3	kJ/kg	230	230	245	
Evaporator inlet	h_4	kJ/kg	230	230	245	
<u>Evaporator Power</u>						
Heat power emitted by heater	Q_{ee}	W	672	673	682	
Heat power by R134a	Q_{er}	W	1800	1800`	1925	
<u>Condenser Power</u>						
Heat Power absorbed by water	Q_{cw}	W	731.5	664.6	480.7	
Heat Power emitted by R134a	Q_{cr}	W	2250	2150	2365	
<u>Compressor</u>						
Fluid power (isentropic)	W_{fi}	W	250	250	300	
Fluid power (actual)	W_{fa}	W	450	350	400	
Compressor isentropic efficiency	η_{cs}	%	55.56	71.43	75.0	
Compressor actual efficiency	η_c	%	37.3	37.2	36.7	
<u>Motor</u>						
Motor speed	N_m	rev/min	1677.06	1675.08	1671.12	
Motor input power	W_{me}	W	470	486	502	
Shaft power output	W_{ms}	W	176	186	175	

Motor efficiency	η_m	%	37.4	38.3	34.9	
<u>Saturation Temperatures</u>						
Evaporator saturation temperature	T_e	$^{\circ}\text{C}$	-10	-10	-10	
Condenser saturation temperature	T_c	$^{\circ}\text{C}$	24	25	31	
<u>COPs</u>						
Overall			3.83	3.70	3.83	
Based on shaft power			4.00	5.14	4.81	
Fluid cycle, isentropic compression			7.20	7.20	6.42	
Carnot cycle based on T_e and T_c			7.74	7.51	6.41	
Carnot cycle based on T_e and T_s			8.77	8.77	8.65	

Appendix B (Pump systems Lab)

B.1 Table of Collected/Observed/Supplied data

Constant operation (3000 rev/min)

	Pump 1			Pressure			
	Torque	Speed	Power	ΔP_1	P_2	P_3	P_4
	(Nm)	(rpm)	(W)	(bar)	(bar)	(bar)	(bar)
1	1.26	3000	397	0.23	-0.30	0.22	0.19
2	1.23	3001	389	0.21	-0.27	0.36	0.34
3	1.19	3003	371	0.19	-0.23	0.52	0.49
4	1.14	3000	362	0.16	-0.20	0.66	0.64
5	1.11	3002	346	0.14	-0.17	0.81	0.79
6	1.04	3005	329	0.11	-0.14	0.96	0.94
7	0.95	2998	300	0.09	-0.10	1.11	1.09
8	0.88	3003	277	0.06	-0.07	1.25	1.24

9	0.77	3001	240	0.04	-0.02	1.4	1.39
10	0.64	3001	205	0.02	-0.01	1.55	1.54

Parallel pump operation

	Pump 1			Pump 2			Pressure			
	Torque	Speed	Power	Torque	Speed	Power	ΔP_1	P_2	P_3	P_4
	(Nm)	(rpm)	(W)	(Nm)	(rpm)	(W)	(bar)	(bar)	(bar)	(bar)
1	1.13	3000	353	1.25	3005	392	0.61	-0.19	-0.21	0.50
2	1.07	3004	335	1.21	3000	373	0.50	-0.15	-0.17	0.71
3	0.99	3000	307	1.12	3005	353	0.39	-0.11	-0.13	0.91
4	0.90	3002	284	1.05	3000	321	0.26	-0.08	-0.09	1.12
5	0.79	3003	245	0.87	2999	277	0.12	-0.04	-0.03	1.33
6	0.61	2998	189	0.62	3002	207	0.03	-0.02	0.0	1.55

Series pump operation

	Pump 1			Pump 2			Pressure			
	Torque	Speed	Power	Torque	Speed	Power	ΔP_1	P_2	P_3	P_4
	(Nm)	(rpm)	(W)	(Nm)	(rpm)	(W)	(bar)	(bar)	(bar)	(bar)
1	1.32	3001	418	1.39	3003	433	0.28	-0.34	-0.17	0.23
2	1.29	2999	404	1.33	2999	418	0.22	-0.28	0.14	0.77
3	1.24	2996	387	1.27	3000	396	0.17	-0.22	0.44	1.28
4	1.14	3000	356	1.15	2998	375	0.12	-0.15	0.77	1.81
5	1.00	2997	312	1.05	2999	327	0.07	-0.09	1.09	2.32
6	0.74	2998	235	0.8	3001	239	0.01	-0.01	1.42	2.86

Suction performance test and cavitation demonstration

	Pump 1			Pressure				Bubbles formation (notes)
	Torque	Speed	Power	ΔP_1	P_2	P_3	P_4	
	(Nm)	(rpm)	(W)	(bar)	(bar)	(bar)	(bar)	
1	1.28	3001	400	0.21	-0.27	0.21	0.18	No bubbles
2	1.27	3001	395	0.19	-0.43	0.18	0.16	Tiny bubbles
3	1.20	2999	372	0.15	-0.65	0.15	0.12	Small bubbles
4	1.09	2999	340	0.09	-0.89	0.10	0.08	Medium bubbles
5	0.98	3000	310	0.06	-0.89	0.06	0.06	Large bubbles
6	0.71	3002	220	0.01	-0.82	0.01	0.0	Very large bubbles

B.2 Sample Calculation

$$\begin{aligned}
 \text{Total head (H)} &= \frac{P_{out} - P_{in}}{\rho g} = \frac{(0.19 \times 100,000) - (-0.27 \times 100,000)}{1000 \times 9.81} \\
 &= 4.99 \\
 P_{out} &= P_4 \\
 P_{in} &= P_2 \text{ for pump 1 / venturi} \\
 &= \frac{P_2 + P_3}{2} \text{ for parallel} \\
 \text{Flow rate (Q)} &= C_d A_1 \left(\frac{2 \Delta P_1}{\rho \left(\left(\frac{D_1}{D_2} \right)^4 - 1 \right)} \right)^{\frac{1}{2}} \\
 &= (0.97)(5.81 \times 10^{-4}) \left[\frac{2(0.19 \times 100,000)}{(1000) \left(\left(\frac{0.0271}{0.0076} \right)^4 - 1 \right)} \right]^{\frac{1}{2}} \\
 &= 1.99 \\
 \text{Pump efficiency (}\eta\text{)} &= \frac{\rho H Q}{\left(\frac{2 \pi N T}{60} \right)} = \frac{(1000)(4.99)(1.99)}{\left(\frac{2 \pi (3000)(1.26)}{60} \right)} \\
 &= 24.9 \\
 \text{Fluid power (W}_F\text{)} &= \rho g H Q = (1000)(9.81)(4.99)(1.99) \\
 &= 97.7 \\
 \text{Shaft power (W}_s\text{)} &= \omega T = \frac{2 \pi N T}{60} = \frac{2 \pi (3000)(1.26)}{60} = 392.7 \\
 \text{Pump efficiency (}\eta\text{)} &= \frac{W_F}{W_s} = \frac{97.7}{392.7} = 0.2487 \dots \\
 &= 24.9\%
 \end{aligned}$$

B.3 Table of Calculated Results**Constant operation (3000 rev/min)**

	Calculated results				
	Total head	Flow rate	Fluid power	Shaft power	Pump efficiency
	H	KQ	W_F	W_s	
	m	km^2/s	W	W	%
1	4.99	1.99	97.7	392.7	24.9
2	6.22	1.907	116.4	373.8	31.1
3	7.34	1.814	130.6	373.8	34.9
4	8.56	1.665	139.8	358.1	39.0
5	9.79	1.557	149.5	348.7	42.9
6	11.01	1.380	149.1	326.7	45.6
7	12.13	1.249	148.6	290.5	49.8
8	13.35	1.019	1.335	276.5	48.3
9	14.58	1.249	178.6	241.9	73.8
10	15.90	0.589	91.9	201.1	45.7

Parallel pump operation

	Calculated results				
	Total head	Flow rate	Fluid power	Shaft power	Pump efficiency
	H	KQ	W_F	W_s	
	m	km^2/s	W	W	%
1	7.03	3.407	235.0	747.7	31.4
2	8.77	2.943	253.2	719.4	35.2
3	10.40	2.599	265.2	662.9	40
4	13.15	2.122	273.7	612.6	44.7
5	13.97	1.442	197.6	521.5	37.9
6	16.82	0.721	119.0	389.6	30.5

Series pump operation

	Calculated results				
	Total head	Flow rate	Fluid power	Shaft power	Pump efficiency
	H	KQ	W_F	W_s	
	m	km^2/s	W	W	%
1	4.99	2.202	107.8	851.4	12.7
2	8.46	2.163	179.5	823.1	21.8
3	11.88	1.716	200.0	785.4	25.5
4	15.34	1.442	217.0	732.0	29.6
5	18.65	1.101	201.4	634.6	31.7
6	22.27	0.416	90.9	438.8	18.8

Suction performance test and cavitation demonstration

	Calculated results				
	Total head	Flow rate	Fluid power	Shaft power	Pump efficiency
	H	KQ	W_F	W_s	
	m	km^2/s	W	W	%
1	4.59	1.907	85.9	402.1	21.4
2	6.01	1.814	106.9	399.0	26.8
3	7.95	1.612	125.7	373.8	33.6
4	9.89	1.249	121.2	339.3	35.7
5	9.58	1.019	95.8	307.9	31.1
6	8.36	0.589	48.3	254.5	19.0

Appendix C (Heat transfer Lab)**C.1 Table of Collected/Observed/Supplied data****Table 1**

Heat transfer surface: pinned surface. Power = 50W							
Air velocity (m/s)	Inlet temperature	Surface temperature	Average exit temperature			Temperature differences	
	T ₁	T ₂	T ₃			ΔT ₁₂	ΔT ₁₃
1	19.0	63.3	26.6	25.2	23.5	44.3	6.1
			25.1				
1.5	19.6	62.0	26.5	24.4	22.2	42.4	4.8
			24.4				
2	18.8	57.5	24.5	22.7	21.0	38.7	3.9
			22.7				
2.5	18.3	54.2	23.3	21.7	20.3	35.9	3.4
			21.7				

Table 2

Heat transfer surface: pinned surface. Power = 50W							
Air velocity (m/s)	Inlet temperature	Surface temperature	Average exit temperature			Temperature differences	
	T ₁	T ₂	T ₃			ΔT ₁₂	ΔT ₁₃
1	19.9	86.4	23.2	23.9	27.8	66.5	5.07
			24.97				
1.5	19.5	88.5	26.7	21.7	20.8	69.0	3.57
			23.07				
2	19.2	82.5	25.5	21.0	19.9	63.3	1.93
			21.13				
2.5	19.4	75.9	24.3	20.8	19.7	56.5	2.2
			21.6				

C.2 Sample Calculation

$$\Delta T_{12} = T_2 - T_1 = 86.4 - 19.9 = 66.5$$